When zero causal Shapley attribution implies a causal null

The causal null in [1, Definitions 2.9–2.10] concerns invariance of the entire outcome law under a single intervention. The game in [1, §4.1.0.2] uses interventional expectations. Lemma 5's prose assumes that every coalition marginal vanishes. Its displayed premise states only equality of single-intervention expectations. That equality alone does not imply zero attribution: the two-node example below with $c = (-1, 1)$ is a counterexample. The stronger prose premise is sufficient. The discussion in [1, §4.1.0.2] notes the absence of a general condition excluding cancellation. Here we characterize when population-wide zero attribution forces single-intervention law invariance, and when it forces every coalition marginal to vanish, in a linear model.

Model and normalization. Let $N = \{1, \dots, n\}$ index all observed features. Their known structural equations and the response are

$$X = BX + \varepsilon, \qquad Y = c^\top X + \eta, \qquad f(x) = c^\top x.$$

The matrix $B \in \mathbb{R}^{n \times n}$ has an acyclic directed support graph: $u \to v$ means $B_{vu} \neq 0$. Thus $B^n = 0$; $I = B^0$ is the identity matrix. The errors $\varepsilon_1, \dots, \varepsilon_n, \eta$ are mutually independent, centered, and have finite variances; each $\mathrm{Var}(\varepsilon_i) > 0$. No Gaussian or continuous-noise assumption is needed. Intervention replaces the equations at the intervened vertices by their assigned values. All n features are players. For $S \subseteq N$, set

$$\begin{aligned} v(S; x) &= \mathbb{E}[f(X) \mid \mathrm{do}(X_S = x_S)], \\ \phi_j(x) &= \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(n - |S| - 1)!}{n!} \, [v(S \cup \{j\}; x) - v(S; x)]. \end{aligned} \tag{1}$$

This is the symmetric game of [1, §4.1.0.2]. That paper uses $p = n$ features and weight $[p\binom{p-1}{|S|}]^{-1}$, equal to the factorial weight in (1). Independent centered η also permits Y in place of $f(X)$. There is no variance rescaling. A *global zero* means $\phi_j(X) = 0$ almost surely. A *single-intervention null* means that $\mathcal{L}(Y \mid \mathrm{do}(X_j = z))$ is the same for every $z \in \mathrm{supp}(X_j)$. We take this support as the intervention domain; [1, Definitions 2.9–2.10] uses a feature domain $\mathcal{X}_j$. In this model the result is unchanged for any real value set containing at least two points, including any larger domain containing the support. A *coalition null* means $v(S \cup \{j\}; X) - v(S; X) = 0$ almost surely for every $S \subseteq N \setminus \{j\}$.

Write e_j for coordinate vector j, and define the weighted heights

$$h_- = \max\{r \geq 0 : e_j^\top B^r \neq 0\}, \qquad h_+ = \max\{r \geq 0 : B^r e_j \neq 0\}.$$

Both sets contain 0. Let D count the descendants of j, including j. Let d and L be the longest directed path lengths ending and starting at j, with length measured in edges.

Theorem 1. Fix B and j, and allow every $c \in \mathbb{R}^n$. The linear map from c to the coefficient vector of ϕ_j has rank $\min(h_- + 1, h_+ + 1)$. The implication

$$\text{global zero} \implies \text{single-intervention null} \quad \text{holds for every } c \quad \iff \quad h_- \geq h_+.$$

The implication from global zero to the coalition null holds for every c if and only if there is a directed path starting at j through all its D descendants and $h_- \geq D - 1$.

If all edge weights are positive, or for Lebesgue-almost-every assignment of nonzero real edge weights on a fixed support, $h_- = d$ and $h_+ = L$. Thus the first criterion becomes $d \geq L$ under those qualifications. The second becomes a path from j through all its descendants and $d \geq D - 1$ under the same qualifications.

Path identity and rank. For a real variable t, define finite polynomials

$$R_c(t) = \sum_{r=0}^{n-1} c^\top B^r e_j \, t^r, \qquad P_a(t) = \sum_{s=1}^{n-1} (B^s)_{ja} t^{s-1} \quad (a \neq j).$$

We first establish

$$\phi_j(x) = x_j \int_0^1 R_c(t)\, dt - \sum_{a \neq j} x_a \int_0^1 (1-t) P_a(t) R_c(t)\, dt. \tag{2}$$

Let B_S be B with rows in S set to zero, and let $x^S \in \mathbb{R}^n$ equal x on S and zero elsewhere. Centering gives

$$\mathbb{E}[X \mid \mathrm{do}(X_S = x_S)] = (I - B_S)^{-1} x^S = \sum_{r=0}^{n-1} B_S^r x^S.$$

This finite series expands the mean along directed paths. A path from an intervened vertex contributes its edge product times the assigned value if none of its later vertices is intervened on. Interpret (1) as averaging over uniform random player orders. A path starting at j with r edges appears when j precedes its other r vertices, with probability $1/(r+1)$. A path starting at $a \neq j$, passing through j after s edges, and then continuing for r edges disappears when a is first and j second among its $s + r + 1$ vertices. These are first and second in relative order on this path; off-path vertices may occur anywhere. The probability is $1/[(s+r)(s+r+1)]$.

The prefix and suffix through j have no other common vertex, since the graph has no directed cycle. Their path sums therefore multiply, including when paths reconverge or weights have mixed signs. The identities

$$\int_0^1 t^r \, dt = \frac{1}{r+1}, \qquad \int_0^1 (1-t) t^{s+r-1} \, dt = \frac{1}{(s+r)(s+r+1)}$$

give (2) after summing the path contributions.

Let $\mathcal{P}_k$ denote the real polynomials of degree at most k, with $\mathcal{P}_{-1} = \{0\}$. Consider the vectors $e_j^\top B^r$ for $0 \leq r \leq h_-$. They are independent. Indeed, in a nonzero dependence choose the least exponent q with a nonzero coefficient and multiply on the right by $B^{h_- - q}$. Only the nonzero multiple of $e_j^\top B^{h_-}$ remains. The same argument applies to the columns $B^r e_j$. It follows that R_c ranges over all of $\mathcal{P}_{h_+}$ as c varies. The independent rows $e_j^\top B^s$, $1 \leq s \leq h_-$, vanish in column j. Their remaining columns therefore span $\mathbb{R}^{h_-}$. These columns are the coefficient vectors of P_a, $a \neq j$, so the P_a span $\mathcal{P}_{h_- - 1}$. Hence

$$\mathrm{span}\{1, (1-t)P_a(t) : a \neq j\} = \mathcal{P}_{h_-}.$$

For the last equality, use $p(t) = p(1) + (1-t)q(t)$ with $q \in \mathcal{P}_{h_- - 1}$, also covering $h_- = 0$.

Equation (2) pairs $R_c \in \mathcal{P}_{h_+}$ with $\mathcal{P}_{h_-}$ under $\langle p, q \rangle = \int_0^1 p(t) q(t)\, dt$. This pairing has rank $\min(h_- + 1, h_+ + 1)$: the spaces are nested and the inner product is positive definite on the smaller space. This proves the rank assertion.

Finally,

$$\mathrm{Cov}(X) = (I - B)^{-1} \, \mathrm{diag}(\mathrm{Var}\, \varepsilon_i) (I - B)^{-\top}$$

is positive definite. Every ϕ_j is linear and centered. Thus $\phi_j(X) = 0$ almost surely if and only if its entire coefficient vector is zero. Equivalently, R_c is orthogonal to $\mathcal{P}_{h_-}$. This step needs no density or full-dimensional support.

Single interventions and sharpness. Solving the structural equations after intervening only at j gives

$$Y \mid \mathrm{do}(X_j = z) = \tau_j z + Z_j, \qquad \tau_j = c^\top (I - B)^{-1} e_j = R_c(1),$$

where the law of Z_j does not depend on z. The variance of X_j is positive, so its support contains two distinct values. If $\tau_j \neq 0$, the intervened means at any two distinct real values differ. If $\tau_j = 0$, the intervened laws coincide. Thus the single-intervention null is exactly $R_c(1) = 0$. This uses equality of laws and does not require a finite supremum of their distances on unbounded supports.

If $h_- \geq h_+$, orthogonality to $\mathcal{P}_{h_-}$ forces $R_c = 0$. For the converse, suppose $h_- < h_+$ and set $m = h_- + 1$. The shifted Legendre polynomial

$$\ell_m(t) = \frac{1}{m!} \frac{d^m}{dt^m} \{t^m (t-1)^m\}$$

has degree m, satisfies $\ell_m(1) = 1$, and is orthogonal to $\mathcal{P}_{m-1}$. The orthogonality follows by integrating by parts m times; all boundary terms vanish and the mth derivative of a polynomial of degree less than m is zero. Since $m \leq h_+$, some c realizes $R_c = \ell_m$. It has global zero attribution and $\tau_j = 1$. The coefficient vector c is chosen using B. This proves the sharp uniform implication, with unit single-intervention effect as the witness normalization.

All coalition marginals. The coalition null is equivalent to $c_k = 0$ at every descendant k of j. One direction follows because both paths starting at j and paths through j blocked by the intervention end at descendants of j. For the other, if $c_j \neq 0$, clamp all other vertices; the marginal has coefficient c_j on x_j. Otherwise choose a descendant k with $c_k \neq 0$ at the smallest directed distance from j, and retain a shortest path P from j to k. Take $S = N \setminus V(P)$, where $V(P)$ is its vertex set, and clamp at x_S. All interior response coefficients on the path vanish by the choice of k. There is no forward shortcut on a shortest path, and no backward edge in an acyclic graph. The marginal's coefficient on x_j is therefore $c_k \prod_{u \to v \text{ in } P} B_{vu} \neq 0$. It cannot vanish almost surely, by the positive-definite covariance of X. This also handles discrete support without assuming that zero is an allowed intervention value.

Attribution depends only on the D descendant response coefficients. The implication to the coalition null is therefore injectivity on those coordinates, or

$$D = \min(h_- + 1, h_+ + 1).$$

Always $h_+ \leq L \leq D - 1$. Equality $h_+ = D - 1$ forces a path through all D descendants. Conversely, such a path imposes a unique vertex order on this acyclic descendant graph. No distinct path of that length can cancel its edge product. Hence it gives $h_+ = D - 1$. The remaining condition is $h_- \geq D - 1$.

For positive weights no path sums cancel, giving $h_- = d$ and $h_+ = L$. For arbitrary nonzero real weights, an entry of B^d associated with a longest incoming path is a nonzero polynomial in the edge weights: it contains that path's monomial. Its zero set is a proper algebraic set of Lebesgue measure zero. Apply the same argument to an outgoing path. This proves the generic qualification and completes Theorem 1.

Examples and limits. For $X_1 = \varepsilon_1$, $X_2 = X_1 + \varepsilon_2$, the attribution is $\phi_1(x) = (c_1 + c_2/2)x_1$ and the single-intervention effect is $\tau_1 = c_1 + c_2$. Thus $c = (-1, 2)$ gives global zero and $\tau_1 = 1$. Conversely, $c = (-1, 1)$ gives a single-intervention null but $\phi_1(x) = -x_1/2$. A single-intervention null alone need not imply zero attribution. Lemma 5's coalition premise remains essential.

The coalition distinction persists even when attribution is zero. For the fork $X_1 = \varepsilon_1$, $X_2 = X_1 + \varepsilon_2$, $X_3 = X_1 + \varepsilon_3$, take $c = (0, 1, -1)$. Then $\phi_1 = 0$ and $\tau_1 = 0$, but the coalition $S = \{2\}$ has $v(S \cup \{1\}; x) - v(S; x) = -x_1$.

Graph lengths alone do not suffice for every signed weight assignment. Take edges $1 \to 2$, $1 \to 3$, $2 \to 4$, $3 \to 4$, $4 \to 5$, $5 \to 6$. All weights are 1 except that $3 \to 4$ has weight -1. At $j = 4$, $d = L = 2$, but $h_- = 1 < h_+ = 2$ because the two incoming length-two path products cancel. The response $c = (0, 0, 0, 1, -6, 6)$ has $R_c(t) = 1 - 6t + 6t^2$, global zero attribution, and $\tau_4 = 1$. The weighted heights are needed at such exceptional weights.

These statements concern an exactly known, fully observed linear graph and exact population-wide zeros, uniformly over unrestricted linear responses. A zero at one observation does not meet the premise. Nor does a signed average: $\mathbb{E}\phi_j(X) = 0$ automatically in this centered model. For fixed B, j, the global-zero coefficient vectors form a proper linear subspace of codimension $\min(h_- + 1, h_+ + 1)$. Thus zero-but-non-null responses have probability zero under any absolutely continuous distribution on unrestricted c. The uniform criteria concern worst cases. No empirical frequency of cancellation in fitted predictors is asserted. The effect τ_j is already directly computable from B, c. No necessary-and-sufficient criterion tailored to restricted response classes is given, nor any bound from near-zero attribution. The sufficient implications still hold on every subclass. The result gives no finite-sample inference guarantee and does not account for graph estimation, hidden variables, or nonlinear mechanisms or asymmetric games. No empirical gain, fitted model, or empirical holdout is involved.

The proof uses standard nilpotent Krylov independence, polynomial Gram forms, and Legendre orthogonality. The causal game and two-variable decomposition are already present in Heskes et al. [2, §2; §4, Figure 1]; exact general structural-model computation is studied by Witter et al. [3, §3, Proposition 3.2]. The result here is the weighted-height rank formula and the characterization of the two uniform implications. No priority or algorithmic improvement is asserted here.

References

[1] G. Raskutti, K. Sankaran, and J. Ye. *Null importance: Disentangling relevance for interpretable machine learning.* arXiv:2609.19511v1 (2026). Definitions 2.9–2.10; §4.1.0.2; Lemma 5 and its following discussion.

[2] T. Heskes, E. Sijben, I. G. Bucur, and T. Claassen. Causal Shapley Values: Exploiting Causal Knowledge to Explain Individual Predictions of Complex Models. *NeurIPS* 33 (2020), 4778–4789.

[3] R. T. Witter et al. *Exactly Computing do-Shapley Values.* arXiv:2602.07203v1 (2026).

Checker

Run `python3 -I check_kris.py` with Python 3.9 or later; only the standard library is used. It checks the examples, including the six-node signed graph and its relabeling, and Legendre degrees 1–8. It checks the path formula, rank, and both null criteria on all ordered signed-unit graphs through four vertices: 760 graphs and 3004 feature checks. It also checks 18 fixed graphs on five to seven vertices with nonzero weights in $\{-2, -1, 1, 2\}$ and non-topological labels: 108 feature checks. These exact finite checks do not replace the proof. SHA-256 of `check_kris.py`:

`4a53296df61cc8787c6b602eea36d4745b32ee3d55ef3d10bf64aa8595b0f1c8`